

RESEARCH INTERESTS

AI Security & Privacy; Data Privacy; Trustworthy Machine Learning.

EDUCATION

Purdue University Ph.D. in Computer Science (Advisor: Ninghui Li)	2023 – 2029 (Expected) USA
Zhejiang University M.E. in Computer Technology. <i>Rank 1/49.</i>	2020 – 2023 China
East China Normal University B.E. in Data Science. <i>GPA 3.61/4 (rank 1/23).</i>	2016 – 2020 China

SELECTED HONORS AND AWARDS

National Winner, Innovation Bowl (Purdue Team Lead), covered by Purdue CS News	2026
NDSS Fellowship , Internet Society (1 of 24 worldwide)	2026
Ross Fellowship , Purdue University	2023
Herbold Scholarship , Purdue University (1 of 7)	2023
Presidential Doctoral Excellence Award , Purdue University (1 of 150)	2023
Excellent Master's Dissertation , China (1 of 43)	2023
Provincial Outstanding Graduate , Zhejiang, China	2023
National Scholarship , China (0.1%)	2022
National Scholarship , China (0.1%)	2021

WORK EXPERIENCE

Research Intern, Perplexity AI Mentor: Dr. Ninghui Li	2026.05 – Present San Francisco, USA
Research Intern, Alibaba Qwen Mentor: Dr. Bolin Ding	2024.05 – 08 Seattle, USA
Research Intern, Microsoft Research Asia Mentor: Dr. Xing Xie . Excellent Internship Award (top 10%)	2022 – 2023 Beijing, China

PUBLICATIONS

First/Co-First	11 publications at CCS (×2), USENIX Security (×2), NDSS, ACL, VLDB, WWW, SIGIR, TKDE, and IEEE Data Eng. Bulletin.
Mentoring	Students I mentored have 5 papers at USENIX Security, CCS poster, SIGIR, and ICDE (×2).

*equal contribution, †mentored student

CCS 2026	Privacy Leakage from a Thousand Words: Millipixel Location Recovery from Dot Maps. Yuntao Du [*] , Tanishq Pauskar ^{†*} , Hao Wang, Jing Su, Ninghui Li. <i>In ACM Conference on Computer and Communications Security (CCS), 2026.</i> ► Achieves meter-level location reconstruction error on national-scale dot maps; working with the U.S. CDC to revise its cartographic guidelines.
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- ACL 2026 **Automated Profile Inference with Language Model Agents.**
Yuntao Du, Zitao Li, Bolin Ding, Yaliang Li, Hanshen Xiao, Jingren Zhou, Ninghui Li.
In Findings of the Association for Computational Linguistics (ACL), 2026.
► First study showing that LLM agents enable automated doxing at web scale.
- USENIX 2026 **Imitative Membership Inference Attack.**
Yuntao Du, Yuetian Chen, Hanshen Xiao, Bruno Ribeiro, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2026.
► A new shadow training paradigm that reduces privacy auditing computation by more than 20×.
- USENIX 2026 **Membership Inference Attacks on Tokenizers of Large Language Models.**
Meng Tong[†], Yuntao Du^{*}, Kejiang Chen, Weiming Zhang, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2026.
► First method to audit privacy leakage of pretraining data through LLM tokenizers.
- USENIX 2026 **Window-based Membership Inference Attacks Against Fine-tuned Large Language Models.**
Yuetian Chen, Yuntao Du, Kaiyuan Zhang, Ashish Kundu, Charles Fleming, Bruno Ribeiro, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2026.
- ICLR 2026 **Membership Inference Attacks Against Fine-tuned Diffusion Language Models.**
Yuetian Chen, Kaiyuan Zhang, Yuntao Du, Edoardo Stoppa, Charles Fleming, Ashish Kundu, Bruno Ribeiro, Ninghui Li.
In International Conference on Learning Representations (ICLR), 2026.
- NDSS 2026 **Cascading and Proxy Membership Inference Attacks.**
Yuntao Du, Jiacheng Li, Yuetian Chen, Kaiyuan Zhang, Zhizhen Yuan, Hanshen Xiao, Bruno Ribeiro, Ninghui Li.
In Network and Distributed System Security Symposium (NDSS), 2026.
► Reformulates MIA as a security game and is the first to exploit membership dependencies for privacy auditing.
- CCS 2025 **Systematic Assessment of Tabular Data Synthesis.**
Yuntao Du, Ninghui Li.
In ACM Conference on Computer and Communications Security (CCS), 2025.
► Identifies fundamental flaws in widely used privacy metrics and proposes a unified evaluation framework.
- IEEE Data Eng. Bulletin **Beyond Data Privacy: New Privacy Risks for Large Language Models.**
Yuntao Du, Zitao Li, Ninghui Li, Bolin Ding.
In IEEE Computer Society Data Engineering Bulletin, 2025.
► First systematic survey of the emerging privacy risks of LLMs beyond training-data privacy.
- USENIX 2025 **SOFT: Selective Data Obfuscation for Protecting LLM Fine-tuning against Membership Inference Attacks.**
Kaiyuan Zhang, Siyuan Cheng, Hanxi Guo, Yuetian Chen, Zian Su, Shengwei An, Yuntao Du, Charles Fleming, Ashish Kundu, Xiangyu Zhang, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2025.
- ICDE 2024 **Real-Time Trajectory Synthesis with Local Differential Privacy.**
Yujia Hu[†], Yuntao Du, Zhikun Zhang, Ziquan Fang, Lu Chen, Kai Zheng, Yunjun Gao.
In IEEE International Conference on Data Engineering (ICDE), 2024.
- VLDB 2023 **LDPTTrace: Locally Differentially Private Trajectory Synthesis.**
Yuntao Du, Yujia Hu[†], Zhikun Zhang, Ziquan Fang, Lu Chen, Baihua Zheng, Yunjun Gao.
In International Conference on Very Large Data Bases (VLDB), 2023.
- WWW 2023 **Towards Explainable Collaborative Filtering with Taste Clusters Learning.**
Yuntao Du, Jianxun Lian, Jing Yao, Xiting Wang, Mingqi Wu, Lu Chen, Yunjun Gao, Xing Xie.
In ACM Web Conference (WWW), 2023.
► Deployed as the recommendation algorithm for Xbox Gaming; granted as a U.S. patent.

SIGIR 2023	Knowledge-refined Denoising Network for Robust Recommendation. Xinjun Zhu [†] , Yuntao Du , Yuren Mao, Lu Chen, Yujia Hu, Yunjun Gao. <i>In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2023.</i>
ICDE 2023	FLBooster: A Unified and Efficient Platform for Federated Learning Acceleration. Zhihao Zeng [†] , Yuntao Du , Ziquan Fang, Lu Chen, Shiliang Pu, Guodong Chen, Hui Wang, Yunjun Gao. <i>In IEEE International Conference on Data Engineering (ICDE), 2023.</i>
TKDE 2023	Finding Materialized Models for Model Reuse. Minjun Zhao, Lu Chen, Keyu Yang, Yuntao Du , Yunjun Gao. <i>In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.</i>
TKDE 2023	MetaKG: Meta-learning on Knowledge Graph for Cold-start Recommendation. Yuntao Du , Xinjun Zhu [†] , Lu Chen, Ziquan Fang, Yunjun Gao. <i>In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.</i>
KDD 2022	Spatio-Temporal Trajectory Similarity Learning in Road Networks. Ziquan Fang, Yuntao Du , Xinjun Zhu, Danlei Hu, Lu Chen, Yunjun Gao, Christian S. Jensen. <i>In ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2022.</i>
SIGIR 2022	Self-Guided Learning to Denoise for Robust Recommendation. Yunjun Gao, Yuntao Du , Yujia Hu [†] , Lu Chen, Xinjun Zhu, Baihua Zheng. <i>In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.</i>
SIGIR 2022	HAKG: Hierarchy-Aware Knowledge Gated Network for Recommendation. Yuntao Du , Xinjun Zhu [†] , Lu Chen, Baihua Zheng, Yunjun Gao. <i>In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.</i>
ICDE 2021	E²DTC: An End to End Deep Trajectory Clustering Framework via Self-Training. Ziquan Fang, Yuntao Du , Lu Chen, Yujia Hu, Yunjun Gao, Gang Chen. <i>In IEEE International Conference on Data Engineering (ICDE), 2021.</i>
VLDB 2021	MDTP: A Multi-source Deep Traffic Prediction Framework over Spatio-Temporal Trajectory Data. Ziquan Fang, Lu Pan, Lu Chen, Yuntao Du , Yunjun Gao. <i>In International Conference on Very Large Data Bases (VLDB), 2021.</i>

PEER-REVIEWED SHORT PAPERS

CCS 2026	A Preliminary Study of LLM Distillation Inference. Edward Chen [†] , Yuntao Du . <i>In ACM Conference on Computer and Communications Security (CCS), poster paper, 2026.</i>
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PATENTS

U.S. Patent	Explainable Cluster-based Collaborative Filtering. Mingqi Wu, Jianxun Lian, Yuntao Du , Jing Yao, Xiting Wang, Bei Lu, Xing Xie. <i>U.S. Patent No. 12,524,502 B2, 2026.</i>
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TEACHING

<i>Instructor, CS 556: Data Security and Privacy, Purdue University.</i>		Fall 2026
• Online offering with 3 graduate students. • Set up the course: designed the syllabus and assignments, recorded the lecture videos. • Extended the course with new content on AI security and privacy.		
<i>Teaching Assistant, CS 556: Data Security and Privacy, Purdue University.</i>		Spring 2025
• 18 graduate students. • Held office hours, answered students' questions, and graded assignments/exams.		

MENTORING

Edward Chen, high school student, Carmel High School. <i>Publications: 1 CCS 2026 Poster</i>	2026 – Present
Patrick Li, B.S. Computer Science, Stanford University. <i>Publications: 1 manuscript under submission</i>	2025 – Present
Tanishq Pauskar, B.S. Computer Science, Purdue University. <i>Publications: CCS 2026</i>	2025 – Present
Meng Tong, M.S. Computer Science, University of Science and Technology of China. <i>Publications: USENIX Security 2026</i>	2025 – Present
Hillary Yang, high school student, Carmel High School. <i>Now B.S. student at Stanford University.</i> <i>Publications: 1 manuscript on arXiv</i>	2024 – 2025
Yujia Hu, M.S. Computer Science, Zhejiang University. <i>Publications: SIGIR 2022, VLDB 2023, ICDE 2024</i>	2022 – 2024
Zhihao Zeng, M.S. Computer Science, Zhejiang University. <i>Publications: ICDE 2023</i>	2022 – 2023
Xinjun Zhu, M.S. Computer Science, Zhejiang University. <i>Publications: SIGIR 2022, SIGIR 2023, TKDE 2023</i>	2022 – 2023

ACADEMIC SERVICE

CONFERENCE REVIEWING

PC, USENIX Security Symposium	2027
PC, International Conference on Very Large Data Bases (VLDB)	2027
PC, ACM Asia Conference on Computer and Communications Security (AsiaCCS)	2027
PC, Neural Information Processing Systems (NeurIPS)	2026
Poster Committee, IEEE Symposium on Security and Privacy (S&P)	2026
PC, International Conference on Learning Representations (ICLR)	2025 – 2027
PC, International Conference on Artificial Intelligence and Statistics (AISTATS)	2025 – 2026
PC, ACM Web Conference (WWW)	2026
PC, ACM Conference on Data and Application Security and Privacy (CODASPY)	2026
PC, International Conference on Web Search and Data Mining (WSDM)	2026
PC, ACM Conference on Information and Knowledge Management (CIKM)	2024 – 2026
PC, ACM Conference on Research and Development in Information Retrieval (SIGIR)	2023 – 2026
PC, International Joint Conference on Neural Networks (IJCNN)	2026 – 2027
PC, Table Representation Learning Workshop (TRL)	2024 – 2025
PC, AAAI Conference on Artificial Intelligence (AAAI)	2023 – 2027
PC, ACM Conference on Information Retrieval in the Asia Pacific (SIGIR-AP)	2023 – 2024

JOURNAL REVIEWING

Reviewer, ACM Computing Surveys (CSUR)
Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)
Reviewer, ACM Transactions on Privacy and Security (TOPS)
Reviewer, International Journal on Very Large Data Bases (VLDBJ)
Reviewer, ACM Transactions on Recommender Systems (TORS)
Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)
Reviewer, IEEE Transactions on Big Data (TBD)

INVITED TALKS

CS 4501, Data Privacy , <i>Guest lecture</i> , University of Virginia Membership Inference Attacks: Progress and Directions	2026 Apr
CS 593, Trustworthy Machine Learning , <i>Guest lecture</i> , Purdue University Membership Inference Attacks: History, Recent Progress, and Future Directions	2026 Mar
Research Trend Automated Profile Inference with Language Model Agents	2025 Jun
Graduate Research Symposium , Purdue University Automated Profile Inference with Language Model Agents	2025 Mar
CS 556, Data Security and Privacy , <i>Guest lecture</i> , Purdue University Differentially Private Data Synthesis	2025 Mar
AI Time Towards Explainable Collaborative Filtering with Taste Clusters Learning	2023 Nov
Microsoft Research Asia Explainable Clustering and Cluster-based Collaborative Filtering	2023 Jan